THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · HARD LEVEL CHAPTER TEST SET 2

Chapter 5: **Describing Motion Around Us** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The ratio of distances covered in the 1st, 2nd and 3rd seconds from rest with uniform acceleration is: 1

(a) 1 : 1 : 1 **(b)** 1 : 4 : 9 **(c)** 1 : 2 : 3 **(d)** 1 : 3 : 5

2. A body is projected upward and takes 5 s to return. The maximum height reached (g = 10) is: 1

(a) 62.5 m **(b)** 25 m **(c)** 31.25 m **(d)** 125 m

3. Assertion (A): A feather and a coin fall together in a vacuum. Reason (R): In the absence of air resistance, all bodies fall with the same acceleration. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

4. A body covers 30 m in the 3rd second from rest with uniform acceleration. Its acceleration is: 1

(a) 10 m/s squared **(b)** 30 m/s squared **(c)** 12 m/s squared **(d)** 6 m/s squared

5. A car decelerates uniformly from 30 m/s to 10 m/s in 5 s. The distance in this time is: 1

(a) 100 m **(b)** 200 m **(c)** 150 m **(d)** 50 m

6. Assertion (A): The area under a velocity-time graph gives the displacement. Reason (R): Displacement is the product of average velocity and time. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

7. A train 100 m long moving at 36 km/h crosses a bridge in 20 s. The length of the bridge is: 1

(a) 20 m **(b)** 100 m **(c)** 300 m **(d)** 200 m

8. A particle with a velocity of 10 m/s due east gets a velocity of 10 m/s due north after 5 s. The magnitude of its average acceleration is about: 1

(a) 4 m/s squared **(b)** 0 **(c)** 2 m/s squared **(d)** 2.8 m/s squared

9. **Assertion (A):** Acceleration is the rate of change of velocity with time. 1
Reason (R): The acceleration of a body is always in the direction of its motion.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** The speedometer of a car measures its velocity. 1
Reason (R): Velocity has both magnitude and direction.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A cyclist is moving at 6 m/s and increases his speed to 11 m/s while covering 34 m. Find the acceleration and time. 2

12. If the velocity of a body is changing but its speed is constant, what type of motion is it? Give an example. 2

13. A body has displacement given by s = 2t + 3t squared. Find the velocity at t = 2 s. 2

14. Why is the average velocity of a body moving at constant speed around a circle for half a revolution not equal to its speed? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A ball thrown up with 20 m/s reaches the roof of a building in 1 s. Find the height of the roof and the velocity at that point (g = 10). 3

16. The equation of motion s = 10t - 2t squared is given. Find the initial velocity, acceleration and when the body momentarily stops. 3

17. Find the displacement of a body given by its v-t graph: +4 m/s for 5 s then -4 m/s for 5 s. 3

OR

A train starting from rest has acceleration 0.5 m/s squared for 1 minute. Find the speed in km/h and the distance covered. 3

18. Calculate the average speed of the tip of a minute hand of length 10 cm (pi = 3.14) in cm/min and the displacement of the tip in 30 minutes. 3

OR

Explain how the third equation of motion can be derived from the first two. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A car on a highway near Nagpur is moving at 72 km/h. The driver sees a barrier ahead and applies the brakes, producing a uniform retardation of 5 m/s^2. 5
(i) Convert the initial speed to m/s.
(ii) Find the time taken to stop.
(iii) Find the distance covered before the car stops.
(iv) If the barrier is only 30 m from where the brakes are applied, find the speed with which the car hits it.

20. Two stones are thrown vertically upward from the same point with the same speed 20 m/s but 2 s apart. When and where do they meet (g = 10)? 5

OR

A body moves in a straight line with the v-t graph: 0 to 2 s: 0 to 6 m/s; 2 to 5 s: 6 m/s; 5 to 7 s: 6 to -2 m/s. Find the displacement, the distance and the average acceleration over 7 s. 5

END OF QUESTION PAPER — **Total: 40 Marks**